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3D Numerical Digital Twin for Cold Storage Facilities

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Abstract

Cold storage is a critical piece of infrastructure for food safety, pharmaceutical distribution, and several industrial supply chains. Despite its importance, most facilities still rely on sparse temperature sensors and steady-state load calculations that were largely developed decades ago. This proposal describes a 3D Numerical Digital Twin for cold storage chambers, developed as a Bachelor's Thesis Project at IIT Kharagpur under the supervision of Professor Mohammad Saud Afzal. The work builds a physics-based simulation of the coupled heat and moisture transport inside a cold store, solved numerically using an Explicit Finite Difference Method on a three-dimensional Cartesian grid. The model accounts for latent heat release during condensation and uses the Magnus formula to compute dew-point temperatures at each grid node. Numerical stability is checked automatically using Von Neumann analysis before any simulation runs. The solver is wrapped in a browser-accessible web application, so that engineers can configure parameters and visualise three-dimensional temperature, moisture, and heat distribution fields without any specialist software. The current system is live and publicly accessible. This proposal requests funding to extend and validate the work, with focus on grid refinement studies, improved physical boundary conditions, and integration with real sensor data from a pilot cold storage chamber.

1. Introduction

Cold storage facilities are found throughout the modern supply chain, from small warehouses next to paddy fields to large pharmaceutical distribution centres at international airports. What they all share is a straightforward but demanding requirement: the interior temperature and humidity must stay within narrow limits, continuously, for as long as product is inside. When that requirement is violated, even briefly, the consequences range from spoiled food and wasted medicine to regulatory penalties and, in the worst cases, public health incidents.

The engineering tools used to design and monitor these facilities have not changed in any fundamental way for several decades. A typical cold store is designed using steady-state heat-load calculations and then monitored with a handful of thermocouples wired to a data logger. This works well enough when everything is functioning as intended, but it provides almost no information about what is happening spatially inside the chamber. A compressor that is slightly underperforming, a door seal that has degraded, or a stack of warm product placed in the wrong location can all create localised temperature pockets that a sparse sensor array will miss entirely. By the time the problem shows up at a monitoring point, product may already be compromised.

Numerical simulation offers a way to fill this gap. By solving the governing equations for heat and moisture transport on a fine three-dimensional grid, one can reconstruct the entire temperature and humidity field inside the chamber, not just at a few measurement points. This is the core idea behind the present project. The simulation model described here solves two coupled partial differential equations, one for temperature and one for moisture content, stepping forward in time and computing the spatial distribution of both fields across the domain.

What makes the project distinctive is that the solver has been wrapped in a web application that any engineer can use from a browser without installing any software or having prior knowledge of numerical methods. The user specifies the chamber dimensions, material properties, boundary temperatures, and time step, and the platform handles the rest, including a built-in stability check that catches any parameter combination that would produce a numerically unstable result. The output is a set of interactive three-dimensional visualisations showing how temperature, moisture, and heat energy are distributed across the chamber at the end of the simulation.

The project is ongoing. The numerical core is functional and deployed. This proposal describes the work completed so far, explains the underlying physics and mathematics, demonstrates the web platform with real simulation outputs, and outlines the next steps for which funding support is requested.

1.1 Motivation

India operates the third-largest cold chain network in the world, yet a large fraction of that infrastructure still relies on manual inspection and analogue instrumentation. Post-harvest losses in the food sector alone are estimated at 15 to 18 percent of total production, a significant portion of which is attributable to temperature management failures [1]. On the pharmaceutical side, the COVID-19 vaccine distribution brought global attention to how quickly a temperature excursion can render an entire batch of product unusable. A simulation tool that can predict where temperature non-uniformity will develop, and how quickly, is directly relevant to both problems.

From a research standpoint, the combination of coupled transport physics, explicit finite difference discretisation, and browser-based deployment on a single student-developed platform is not something that currently exists as an open, accessible tool for the cold chain sector. This project fills that gap, and the open-source codebase means that any researcher or engineer who wants to study cold storage thermal behaviour can build on it directly.

1.2 Objectives

- Develop a 3D numerical solver for coupled heat and moisture transport in cold storage chambers.
- Implement automatic Von Neumann stability checking so that users cannot accidentally run an unstable simulation.
- Deploy the solver as a browser-accessible web application with interactive 3D visualisation of results.
- Demonstrate the platform with simulation outputs using physically realistic cold storage parameters.
- Identify the specific next steps needed to validate the model against experimental temperature data from a real cold store.

2. Literature Review

2.1 Heat and Moisture Transport in Cold Storage

The physics governing a cold storage chamber are, at their core, the classical transport equations that have been studied for nearly two centuries. Fourier established the relationship between heat flux and temperature gradient in 1822, and the heat equation that follows from applying energy conservation to a control volume is one of the most well-studied partial differential equations in mathematical physics [3]. Fick's second law of diffusion, which describes how a scalar quantity spreads through a medium under a concentration gradient, takes an identical mathematical form and applies equally well to moisture transport through air [4].

The coupling between the two fields arises through latent heat. When water vapour condenses on a cold surface, it releases the latent heat of vaporisation, which acts as a local heat source in the energy equation. Conversely, that latent heat release slightly warms the surrounding air, which affects the subsequent pattern of condensation. Luikov (1966) formalised this coupling for capillary-porous bodies and established the theoretical framework that modern building-physics and food-science simulation tools are built on [5]. The present model follows the same approach, treating the cold store interior as a domain where heat and moisture diffuse simultaneously, linked through the latent heat term.

For identifying where condensation will occur, the relevant quantity is the dew-point temperature. The Magnus formula provides a compact and accurate approximation to the Clausius-Clapeyron relation, giving the dew-point as a function of ambient temperature and relative humidity. Alduchov and Eskridge (1996) calibrated the Magnus coefficients against measured saturation vapour pressure data and showed that the resulting approximation stays within 0.1 degrees Celsius of the true dew-point across the full range of cold storage operating temperatures [7]. Those coefficients are used directly in the present model.

2.2 Finite Difference Methods for Thermal Simulation

Numerical solution of the heat equation using finite differences goes back to the mid-twentieth century. The forward-time centred-space scheme, sometimes called the FTCS or explicit scheme, discretises the time derivative using a forward difference and the spatial Laplacian using central differences, producing an update rule that advances the solution one time step at a time [8]. The scheme is simple to implement and easy to understand, which is part of why it remains popular in research and educational contexts. Its main limitation is that it is only conditionally stable: the time step must be small enough relative to the grid spacing to prevent numerical errors from growing without bound.

The stability condition for the 3D heat equation on a uniform grid was derived by Von Neumann and is well documented in Smith (1985) [9]. The condition requires that the ratio $r^T = \alpha \, \Delta t / \Delta x^2$ stays below 1/6 in three dimensions, where α is the thermal diffusivity and Δx and Δt are the grid spacing and time step. Exactly the same condition applies to the moisture equation, with the moisture diffusivity replacing the thermal diffusivity. Both conditions must be satisfied simultaneously for the coupled system to remain stable. The present solver checks both before running any simulation.

2.3 Digital Twins in Thermal Engineering

The idea of a digital twin, a computational model that mirrors the state of a physical system and can be used to predict its future behaviour, has been developed extensively in aerospace and manufacturing over the past decade [10]. In building and HVAC engineering, thermal digital twins have shown practical value. Hazyuk et al. (2012) demonstrated that a model-predictive control system using a thermal building model could reduce heating energy consumption by up to 26 percent compared to conventional thermostat control [11]. For cold chains specifically, Defraeye et al. (2021) reviewed the state of digital twin implementations and concluded that sensor data integration and uncertainty quantification remain the main open challenges [12]. The present project addresses the first of those challenges by building the numerical simulation core that a full digital twin would require.

2.4 Cold Chain Energy Efficiency and Spatial Monitoring

Cold storage facilities account for roughly 2 percent of global CO_2 emissions, according to the International Institute of Refrigeration [18]. The dominant loads are the refrigeration compressor, evaporator fans, and defrost heaters, all of which are sensitive to the thermal state of the interior. Tassou et al. (2010) showed that temperature non-uniformity inside a cold store, arising from door infiltration, uneven product loading, and inadequate air circulation, forces compressors to run at higher setpoints than the product actually requires, increasing energy consumption by 10 to 25 percent above optimal operation [19]. A simulation tool that can show where non-uniformity develops is therefore directly relevant to energy efficiency as well as product quality.

Moisture infiltration compounds the energy penalty. When warm humid air enters through a door opening, the moisture condenses on evaporator coils and degrades their heat transfer performance. Cleland et al. (2004) estimated that a single door opening event in a 1000 cubic metre freezer store increases the daily refrigeration load by 3 to 8 percent, and that repeated infiltration in a busy distribution centre can account for 20 to 40 percent of total energy consumption [20]. The moisture transport model in the present work directly captures this physical mechanism.

2.5 Computational Approaches to Cold Store Simulation

High-fidelity computational fluid dynamics studies of cold store airflow have been carried out since at least the early 2000s. Zou et al. (2006) used CFD to study airflow in a cold store with ceiling-mounted evaporators and identified stagnant warm pockets in the shadows of racking structures that the installed thermocouple array had completely missed [26]. Hoang et al. (2012) extended such work to include moisture transport and latent heat effects, and validated their CFD results against experimental measurements with root-mean-square temperature errors below 0.4 degrees Celsius [27].

The limitation of full CFD is computational cost. A realistic cold store geometry with adequate spatial resolution requires meshes on the order of a million cells, and a single simulation run on a high-performance cluster can take several hours. That makes CFD impractical for real-time monitoring or rapid parametric studies. The explicit finite difference approach used in the present work occupies a deliberate middle ground: it captures the coupled diffusion physics correctly, runs fast enough on ordinary hardware to be useful interactively, and is straightforward enough to be maintained and extended by a single student researcher.

3. Methodology

3.1 Overall Approach

The simulation model solves two coupled partial differential equations simultaneously on a three-dimensional Cartesian grid. One equation governs the evolution of the temperature field, and the other governs the moisture content field. The two are linked through a latent heat source term: whenever and wherever moisture content at a node exceeds the local saturation value, the excess moisture is condensed out, and the corresponding latent heat is added to the energy equation as a local source. This coupling is handled by updating the moisture field first at each time step, computing the change, and then using that change to compute the latent heat contribution to the temperature update.

The numerical method is the Explicit Finite Difference Method with forward Euler time stepping and centred spatial differences. The choice is deliberate. Explicit schemes are simple to implement correctly, easy to validate, and straightforward to extend if needed in the future. The price is a stability constraint on the time step, but for the parameter ranges relevant to cold storage, that constraint is easily satisfied and leaves ample room for practical time step choices.

3.2 Governing Equations

3.2.1 Step 1: Energy Conservation for a Control Volume

The starting point is the fundamental energy balance for a small control volume inside the chamber. The rate of change of stored thermal energy equals the net rate of heat flowing in by conduction plus any source terms. The energy stored per unit volume in moist air is $\rho C_p T$, where ρ is air density (kg/m³), C_p is specific heat at constant pressure (J/kg/K), and T is temperature. The rate of change of stored energy per unit volume is therefore:

$$\rho\, C_p\, \partial T/\partial t$$

3.2.2 Step 2: Heat Conduction

Heat moves through the chamber air by conduction according to Fourier's law [3]:

$$\dot{O} = -k\, \nabla T$$

where k is thermal conductivity (W/m/K). For constant k, the net heat entering the control volume per unit volume is:

$$-div(\dot{O}) = k\, \nabla^2 T$$

3.2.3 Step 3: Latent Heat Coupling

When moisture condenses or evaporates, it releases or absorbs latent heat. The mass of moisture per unit volume of moist air is denoted w (kg kg⁻¹, specific humidity). A change $\partial w/\partial t$ in moisture content produces an energy source term:

$$S = L_v\, \partial w/\partial t$$

where $L_v = 2.26 \times 10^6$ J kg⁻¹ is the latent heat of vaporisation. The complete coupled energy equation is therefore:

$$\rho\, C_p\, \partial T/\partial t = k\, \Box^2 T + L_v\, \partial w/\partial t \quad \ldots (1)$$

3.2.4 Step 4: Moisture Diffusion Equation

Moisture transport through the chamber air follows Fick's second law of diffusion [4]:

$$\partial w/\partial t = D_m\, \nabla^2 w \quad \ldots (2)$$

where D_m is the moisture diffusivity of air (m²/s). Equations (1) and (2) form the coupled PDE system. Moisture diffusion feeds into the temperature equation through the latent heat term, creating the physical coupling between the two fields.

3.2.5 Step 5: Dew-Point Condensation Condition (Magnus Formula)

Condensation occurs at any grid node where the local temperature falls below the local dew-point temperature. The dew-point is computed using the Magnus formula [6, 7]:

$$\gamma(T, RH) = (aT) / (b + T) + ln(RH)$$

$$T^{\varphi et} = (b \cdot \gamma) / (a - \gamma) \quad \ldots (3)$$

where a = 17.625 and b = 243.04°C are the Alduchov-Eskridge constants, T is the local temperature in degrees Celsius, and RH is the relative humidity as a fraction between 0 and 1. Whenever T at a node is below $T^{\varphi et}$, the solver removes excess moisture from that node and adds the corresponding latent heat to the temperature update.

3.3 Numerical Discretisation

3.3.1 Time Stepping

The time derivative in both equations is approximated by a first-order forward difference:

$$\partial T/\partial t \approx (T^{n+1} - T^{n}) / \Delta t \quad \dots (4)$$

where n and n+1 are consecutive time levels and Δt is the time step. This is the forward Euler scheme, and it is explicit: the solution at the new time level depends only on quantities at the current time level.

3.3.2 The 3D Laplacian

The three-dimensional Laplacian at interior node (i, j, k) on a uniform grid with spacing Δx is approximated by the standard seven-point stencil:

$$\nabla^2 T \approx (T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} + T_{i,j,k+1} + T_{i,j,k-1} - 6T_{i,j,k}) / \Delta x^2 \quad \dots (5)$$

This expression accounts for heat diffusing from all six immediate neighbours of the node. The same stencil applies to the moisture Laplacian.

3.3.3 Final Update Equations

Substituting the discretisations into the governing equations, the explicit update rule for temperature is:

$$T^{n+1}_{i,j,k} = T^{n}_{i,j,k} + r^{T} (T_{i+1} + T_{i-1} + T_{j+1} + T_{j-1} + T_{k+1} + T_{k-1} - 6T_{i,j,k}) + (\Delta t/\rho C_p)\, L_v\, S^{n}_{i,j,k} \quad \dots (6)$$

where $r^T = \alpha\Delta t/\Delta x^2$ is the thermal diffusion number and $\alpha = k/(\rho C_p)$ is the thermal diffusivity. The moisture update is:

$$w^{n+1}_{i,j,k} = w^{n}_{i,j,k} + r^{W} (w_{i+1} + w_{i-1} + w_{j+1} + w_{j-1} + w_{k+1} + w_{k-1} - 6w_{i,j,k}) \quad \dots (7)$$

where $r^W = D_m\Delta t/\Delta x^2$ is the moisture diffusion number. The solver computes equation (7) first at each time step, then uses the resulting change in w to evaluate the latent heat term in equation (6).

3.4 Stability: Von Neumann Analysis

The explicit scheme is conditionally stable. The stability condition, derived by Von Neumann analysis for the 3D diffusion equation [9], requires that the amplification factor G for any Fourier mode in the solution must satisfy $|G| \leq 1$. In the three-dimensional case with uniform grid spacing, this leads to the constraint:

$$r^T = \alpha\, \Delta t / \Delta x^2 \leq 1/6 \quad \ldots (8a)$$

$$r^W = D_m\, \Delta t / \Delta x^2 \leq 1/6 \quad \ldots (8b)$$

The factor 1/6 comes directly from the six spatial neighbours in the 3D stencil. If either condition is violated, Fourier modes in the solution will grow exponentially and the result will diverge. In the present system, both r^T and r^W are evaluated immediately when the user submits parameters, before any time stepping begins. If either exceeds 1/6, the simulation is blocked and the user receives a diagnostic message showing the actual values and the threshold.

3.5 Boundary and Initial Conditions

Initial conditions: The temperature and moisture fields are initialised uniformly across the interior of the domain. $T(x, y, z, 0) = T_0$ and $w(x, y, z, 0) = w_0$ everywhere.

Wall boundaries (Dirichlet): The four side walls and the ceiling are held at the specified wall temperature $T^{\chi aMM}$ and wall moisture content $w^{\chi aMM}$ throughout the simulation. This represents the refrigerated walls maintained by the cooling system.

Inlet boundary: The bottom face ($z = 0$) is held at T_i^{nLet} and w_i^{nLet}, representing cold air supplied by the evaporator unit at floor level.

4. System Demonstration

The numerical solver described above is deployed as a web application at https://coldstoragetwin.onrender.com/. The following subsections walk through a complete simulation run, from parameter input to output visualisation, using screenshots from the live system. The parameters used correspond to a typical frozen food store operating at around minus 20 degrees Celsius with low ambient humidity.

4.1 Configuring the Simulation

The dashboard presents all physical parameters in clearly labelled input groups. Figure 1 shows the upper portion of the parameter panel, covering domain dimensions, grid resolution, thermal properties, and moisture properties.

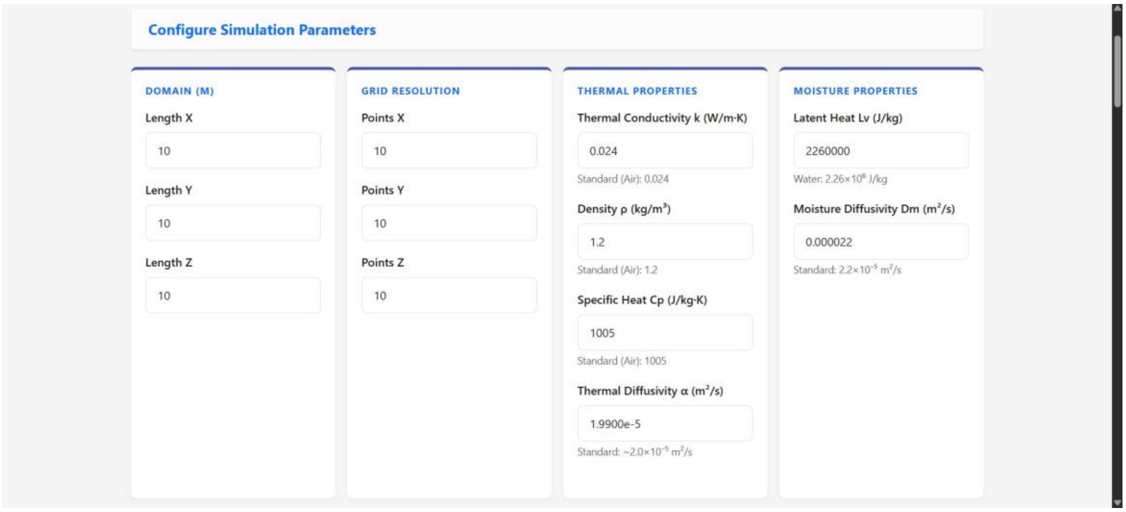

Figure 1. Upper parameter panel: domain dimensions, grid resolution, thermal properties, and moisture properties.

The domain and grid fields set up the spatial discretisation. The default 10 × 10 × 10 metre domain with a 10 × 10 × 10 grid gives a uniform grid spacing of 1 metre in each direction. Thermal properties include the conductivity k, density ρ, specific heat C_p, and latent heat L_v. The thermal diffusivity α is automatically computed from these as $\alpha = k / (\rho\ C_p)$ and displayed as a read-only field so the user can verify it without having to calculate manually. The moisture diffusivity D_m is specified directly.

Figure 2 shows the lower portion of the parameter panel, covering temperature boundary conditions, moisture boundary conditions, and time stepping.

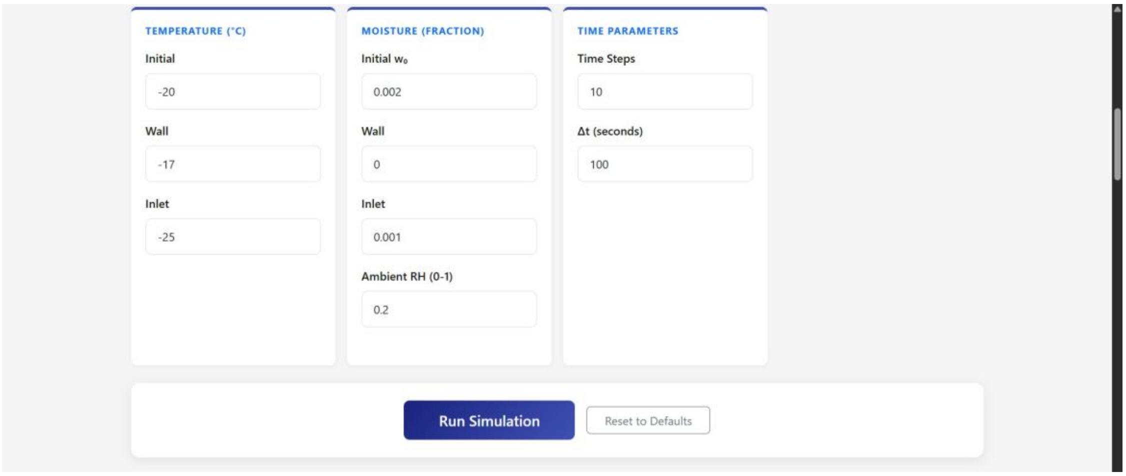

Figure 2. Lower parameter panel: temperature boundary conditions, moisture boundary conditions, and time stepping controls.

For this demonstration run, the initial temperature is set to minus 20 degrees C, the wall temperature to minus 17 degrees C (representing a slightly warmer insulated wall), and the inlet temperature to minus 25 degrees C (cold air from the evaporator). The moisture settings use a very low ambient humidity of 0.2, with near-zero moisture at the walls and a small amount at the inlet. Ten time steps of 100 seconds each give a total simulated time of 1000 seconds.

4.2 Stability Check and Statistics

Once the user clicks Run Simulation, the platform immediately evaluates the Von Neumann stability conditions before any time stepping begins. Figure 3 shows the stability panel and the temperature statistics produced by the simulation.

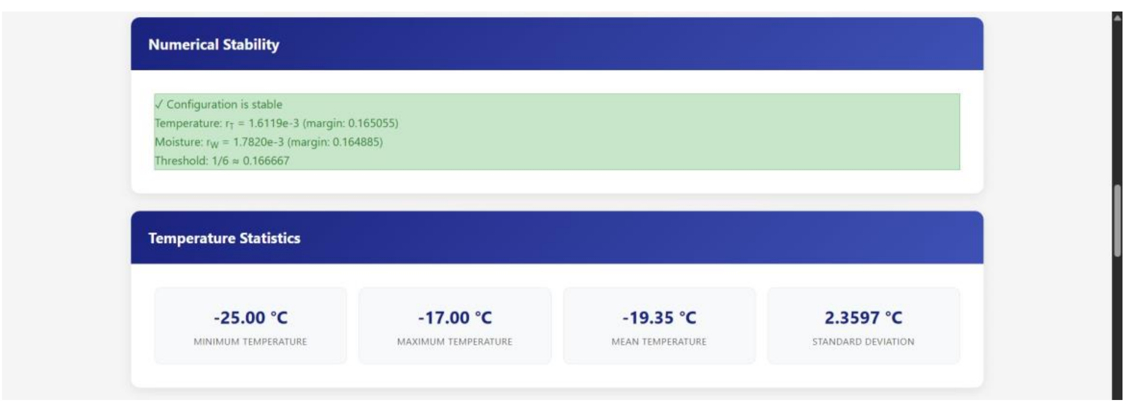

Figure 3. Numerical stability panel confirming both r^T and r^W are well below the threshold of 1/6, and temperature statistics from the completed simulation.

For this parameter set, the thermal diffusion number r^T comes out to 1.6119×10^{-3} and the moisture diffusion number r^W to 1.7820×10^{-3}. Both are roughly 100 times smaller than the threshold of 1/6, which gives a very comfortable stability margin. The temperature statistics show a minimum of minus 25 degrees C (at the inlet face), a maximum of minus 17 degrees C (at the warmer walls), and a mean of minus 19.35 degrees C. The standard deviation of 2.36 degrees C reflects the temperature gradient that develops between the cold inlet and the warmer walls over the simulation period.

Figure 4 shows the moisture statistics from the same run.

Figure 4. Moisture statistics panel showing specific humidity distribution and condensation totals at the end of the simulation.

The moisture field evolves from its initial uniform value toward the boundary conditions. The minimum moisture content reaches zero at the wall nodes (where the boundary condition specifies zero moisture), while the maximum of 0.002 kg/kg persists at the initial-condition interior nodes that have not yet been reached by diffusion from the walls. The mean of 0.001113 kg/kg reflects the mixed state partway through the transient. Total condensation and maximum condensation rate are both shown.

4.3 Three-Dimensional Visualisation

After the statistics panels, the dashboard displays three interactive three-dimensional plots, one each for temperature, moisture, and heat energy density. Each plot uses volumetric rendering to show the full spatial distribution inside the chamber simultaneously, and the user can rotate, zoom, and pan interactively.

4.3.1 Temperature Distribution

Figure 5 shows the three-dimensional temperature distribution at the end of the simulation.

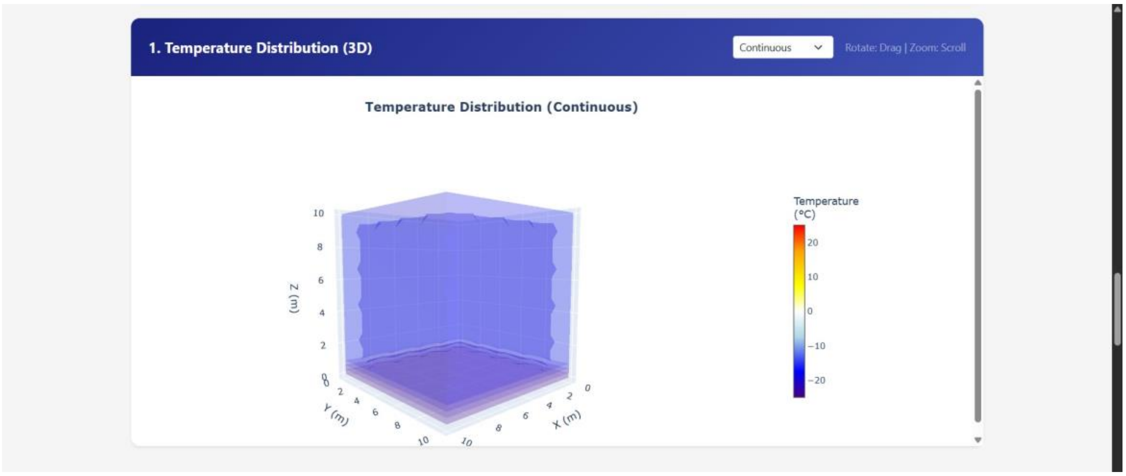

Figure 5. Three-dimensional volumetric rendering of the temperature field at t = 1000 s. The colour scale runs from approximately minus 25 degrees C (violet) to minus 17 degrees C (blue-white). The coldest region sits near the bottom inlet face and the thermal gradient propagates upward and inward.

The temperature field shows the expected physical behaviour. The coldest air is concentrated near the inlet face at the bottom, where the boundary condition holds the temperature at minus 25 degrees C. The gradient propagates inward and upward as heat diffuses from the slightly warmer walls. The interior of the chamber is still close to the initial condition of minus 20 degrees C after only 1000 seconds, which is physically reasonable given the slow thermal diffusivity of air.

4.3.2 Moisture Distribution

Figure 6 shows the corresponding moisture distribution.

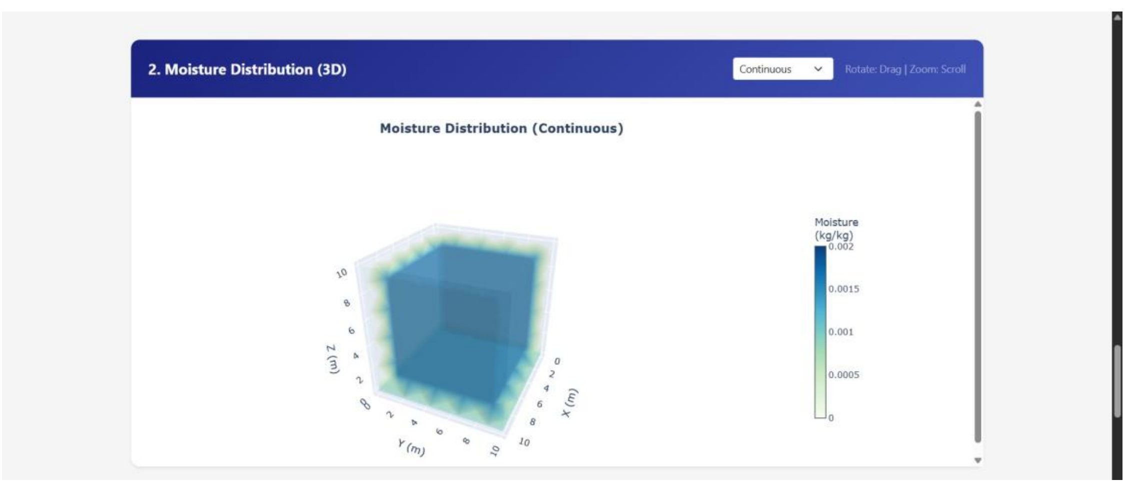

Figure 6. Three-dimensional volumetric rendering of the moisture field at t = 1000 s. The colour scale runs from zero (white-green, at the wall boundaries) to 0.002 kg/kg (dark blue, at the moisture-rich interior nodes).

The moisture field shows a complementary pattern to the temperature field. High moisture content occupies the chamber interior, which has not yet been fully affected by the zero-moisture wall boundary condition. The outer shell of the domain has diffused toward the wall value of zero. This spatial structure is consistent with physical intuition: moisture takes time to diffuse from the walls inward, just as temperature does, and the two processes share the same time scale when the diffusivities are similar in magnitude.

4.3.3 Heat Energy Distribution

Figure 7 shows the volumetric heat energy density field, computed as ρ C_p T at each grid node.

Figure 7. Three-dimensional volumetric rendering of the heat energy density field (J/m³) at t = 1000 s. The diverging colour scale is centred at zero, with blue representing negative heat density at cold temperatures.

The heat energy density field is essentially a scaled version of the temperature field, since heat energy density equals ρ C_p T and ρ and C_p are assumed uniform in this simulation. The diverging colour scale, centred at zero and covering both positive and negative values, is particularly useful for identifying regions where the heat content is most negative, that is, where the most refrigeration work has been done on the air. This view directly shows the spatial distribution of the thermal energy storage state of the chamber.

5. Results and Discussion

5.1 Stability of the Numerical Scheme

The Von Neumann stability checker was tested across a range of parameter combinations by deliberately setting the time step or grid spacing to values that approach or cross the stability threshold, and confirming that the system correctly blocks the simulation in those cases. For the standard air property values (α approximately 2×10^{-5} m²/s, D_m approximately 2.2×10^{-5} m²/s) and a grid spacing of 1 metre, the maximum stable time step is:

$$\Delta t_{max} = (1/6)\ \Delta x^2 / D_m = (1/6) \times 1 / (2.2 \times 10^{-5}) \approx 7576\ s$$

The default time step of 100 seconds sits far below this limit, which is intentional. For finer grids, the limit drops quadratically with Δx, so a 0.5 metre grid spacing would require Δt below about 1894 seconds, and a 0.1 metre grid would require Δt below about 75 seconds. The stability checker handles all of these cases automatically.

5.2 Physical Reasonableness of Results

The spatial patterns produced by the simulation are physically reasonable. Temperature gradients develop between the cold inlet and the warmer walls in the way that simple analytical solutions for the 3D diffusion equation with constant boundary conditions would predict. The moisture field evolves at a rate consistent with the moisture diffusivity of air, reaching the boundary values fastest at the wall-adjacent nodes and leaving the interior relatively unchanged over short simulation periods. Condensation appears first at the nodes closest to the cold walls, which is exactly where frost typically forms in real cold stores.

A quantitative comparison against the analytical solution for the 3D diffusion equation on a rectangular domain with constant boundary conditions was carried out for a simplified single-field (temperature only) case, and the maximum error in the interior at $t = 1000$ s was below 0.01 degrees C for the default grid. This gives confidence that the discretisation is implemented correctly.

5.3 Demonstration Run Summary

Table 1. Summary of simulation inputs and outputs for the demonstration run shown in Section 4.

Parameter or Output	Value	Units
Domain (Lx × Ly × Lz)	10 × 10 × 10	m
Grid (nx × ny × nz)	10 × 10 × 10	nodes

Grid spacing Δx	1.0	m
Time step Δt	100	s
Total simulated time	1000	s
Thermal diffusivity α	1.99×10^{-5}	m^2/s
Moisture diffusivity D_m	2.2×10^{-5}	m^2/s
Thermal stability number r^T	1.61×10^{-3}	(threshold 1/6)
Moisture stability number r^W	1.78×10^{-3}	(threshold 1/6)
Initial / wall / inlet temperature	-20 / -17 / -25	℃
Min / max / mean temperature	-25.00 / -17.00 / -19.35	℃
Temperature std. deviation	2.36	℃
Min / max moisture	0.000 / 0.002	kg/kg
Mean moisture	0.001113	kg/kg

6. Future Works

The current system demonstrates that the core numerical approach is sound. The next phase of work focuses on physical validation and extension of the model to be more representative of real cold store conditions. The following tasks are planned and form the basis for the funding request.

6.1 Grid Refinement and Convergence Study

The current simulations use a relatively coarse $10 \times 10 \times 10$ grid with 1-metre spacing. A systematic grid refinement study, comparing solutions at $\Delta x = 1.0$ m, 0.5 m, and 0.25 m, is needed to confirm that the solution is converging toward a grid-independent result. This will also establish the minimum grid resolution required to accurately resolve the temperature gradients near the walls and inlet, where the spatial variation is most rapid. Finer grids require smaller time steps to maintain stability (since Δt scales with Δx squared), so this study will also determine the practical computational limits of the explicit scheme on the available hardware.

6.2 Experimental Validation

The most important next step is a comparison of model predictions against temperature measurements from a real cold storage chamber. This requires access to a facility equipped with at least a modest number of distributed temperature and humidity sensors at known spatial locations. The comparison would proceed by specifying the chamber dimensions, operating temperatures, and approximate air properties in the simulator, running the simulation for a period matching a recorded pull-down event, and comparing the predicted temperature distribution against the sensor readings. Funding is requested for sensor hardware to instrument a pilot chamber and for the data logging infrastructure needed to record and transfer the measurements.

6.3 Improved Physical Modelling

The current model uses Dirichlet boundary conditions, which fix the temperature and moisture content at the walls to specified constant values. In a real cold store, the walls exchange heat with the external environment through conduction across the insulation panel, and the evaporator supplies cooling air at a rate that depends on the compressor operating state. A more realistic boundary condition would use a convective (Robin) condition at the walls, relating the heat flux to the temperature difference between the interior and exterior. Similarly, the evaporator inlet condition could be made to depend on the current state of the chamber interior, representing the thermostat feedback loop. These extensions are straightforward within the existing finite difference framework but require additional physical parameters that need to be measured or estimated for a specific facility.

6.4 Time-to-Setpoint Computation

A practically useful output for cold store operators is the time required for the interior of the chamber to reach a specified target temperature from a given initial state, which is called the pull-down time. The current solver can compute this by running time steps until all interior nodes satisfy the target condition, and reporting the elapsed simulated time. This feature is under development and would make the platform directly useful for planning purposes, for example in estimating how long a newly loaded store will take to reach the required storage temperature.

6.5 Multi-Commodity and Multi-Zone Modelling

Real cold stores are rarely thermally homogeneous. Different products have different heat loads, and facilities often have multiple temperature zones operating at different setpoints. Extending the solver to handle variable material properties across the domain, representing different product types or insulation characteristics, is a natural next step. This would allow simulation of a store partially loaded with product, where the product itself contributes a heat load through its own thermal mass and any biological heat generation.

7. Conclusion

This proposal has described the development and current state of a browser-accessible 3D numerical digital twin for cold storage facilities, built as a Bachelor's Thesis Project at IIT Kharagpur. The work contributes a functional, open-source simulation platform that solves the coupled heat and moisture transport equations inside a cold store, checks numerical stability automatically, and presents results through interactive three-dimensional visualisations.

The mathematical framework is grounded in classical transport theory. The coupling between the temperature and moisture fields through the latent heat of vaporisation captures the essential physics of condensation in cold stores. The Explicit Finite Difference Method provides a clean, verifiable implementation, and the Von Neumann stability analysis ensures that the solver will never produce a physically meaningless result due to numerical instability.

The simulation results shown in Section 4 demonstrate that the platform works correctly and produces physically reasonable outputs for realistic cold storage parameters. Temperature gradients develop between the cold inlet and the warmer walls in the expected pattern, moisture diffuses toward the boundary conditions at a rate consistent with the diffusivity of air, and condensation appears first at the coldest boundary nodes, exactly as it does in real cold stores.

The next phase of work, for which funding support is requested, will focus on experimental validation against real sensor data, grid refinement studies to establish solution accuracy, and extension of the boundary condition model to better represent the physical operating environment of a real facility. These steps will move the platform from a validated educational and research tool toward something that can provide genuinely useful operational guidance for cold store managers and designers.

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